AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

# TM240
# Introduction to computer graphics

## Final Examination
## Summer Semester 2025/2026

## Time Allowed: 2 Hours

AOU EGYPT

**Instructions:**

- Total Marks: **100**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

الجامعة العربية المفتوحة - مصر
Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt



CamScanner

**Part I:** This part consists of 10 multiple choice questions. For each question, encircle the label a, b, c, or d as correct answer. Each question carries 2 marks.

## Question 1 [20 Marks]

1 ..... function is used to concatenate transformation to the right of the current transformation matrix.

A strcat B preConcatenate
C concatenate D cat

2 The .....colour system is commonly used in printers.

A RGB B CIEXYZ
C CYMK D None of the above

3 to create an image dynamically in computer memory a ..... Object must be associated with the BufferedImage object.

A setPixel B WritableRaster
C raster=bi.getRaster D None of the above

4 Machine dependency is a property of graphics software of .....level

A PHIGS B Hardware
C Operating system D GKS

5 The..... class can be used to manage a printing process .

A printDialog B pageIndex
C pageFormat D PrinterJob

6 The function .....don't clear the existing transformations in the AffineTransform object before applying the new transformation.

A scale B setToScale
C rotate D CreateTransformedShape

7 .....are geometric transformations applied to objects to achieve proper placement of objects in the scene.

A World transformation B Viewing transformation
C Object transformation D modeler

8 the class ..... Is used to geometrically alter a given image.

A RescaleOp B LookupOp
C AffineTransformOp D ConvolveOp

9 .....is the mapping of the object spaces into world space.

A Affine transformation B Object transformation
C Viewing transformation D Generic transformation

10 In the class .....there is no separation between modeling and rendering

A Graphics B J2D
C Graphics2D D None of the above

AOU EGYPT

الجامعة العربية المفتوحة - مصر
Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt



CamScanner

**Part II:** This part consists of four questions with a total of 80 Marks, solve all questions.

**Question 2 [20 Marks]**

a. Write a java code segment to draw the Latin letter



You must use GeneraPath and suitable stroke. [5 marks]

b. Flip the shape in a) vertically to be similar to: [5 marks]

c. Assuming that we have a number x that is represented in 15 bits. Write down appropriate java code segment to divide this number into 3 colors (red, green and blue) with each color component of one byte. [10 marks]

**Question 3 [20 Marks]**

a. Design a custom filter to brighten and sharpen an image in one convolution operation. [4 marks]

b. Having an image object of type BufferedImage that has the following artifacts: [4 marks per point]

1. image contrains objects with blured borders
2. image is sheared towards the positive x-directions
3. image has inverted colors
4. image is zoomed out

Write a java code segment to resolve the above problems.

**Question 4 [20 Marks]**

a. Write a function in java code to implement Mandlbrot fractal sequence with maximum number of iterations of 4096 for an arbitrary setting of the imaginary constant C. [10 marks]

b. using constructive area geometry, develop a java code to output the following: [10 marks]




**Question 5 [20 marks]**

Given the following list of control points:
(0,40)
(38,2)
(76,45)

a. Derive the parametric Bezier equation for the curve controlled by the above point list.
b. Write a code to implement the parametric equations that you derived in the previous point.

End of questions

الجامعة العربية المفتوحة - مصر
Arab Open University
التمويل والتسجيل
الامتحانات
Arab Republic of Egypt



CamScanner